

Consumption & Investment

Recall, $Y = (C) + (I) + G + NX$

- C & I are two important components of the national output

eg, share in national o/p is -

	U.S.A	India
C + I	~ 82%	~ 93%
C	~ 71%	~ 57%
I	~ 11%	~ 36%

- Part of o/p that is not consumed too plays an important role
 - referred to as the savings
 - ↓
 - crucial for funding investment
 - affects the economic performance (especially in the long run)

Savings driven economic growth!

USA - 19th Century

Japan - 20th Century

East Asia - most recently

high growth

high savings relative to income → high investment ↑

low " → low " ↓
low growth

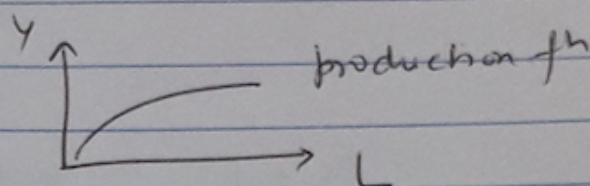
- Consumption growth can also lead to economic growth

Consp. growth $\rightarrow$ high spending $\rightarrow$ high Aggregate Demand

Higher employment $\leftarrow$ Higher O/p $\leftarrow$

[Note that $Y = f(L, K)$]

$\uparrow$ $\uparrow$
high $Y \Rightarrow$ high L



e.g., Consumption driven boom in the USA in the 1990s

Reverse happened in 2007-9 (consp. $\downarrow \rightarrow$ growth $\downarrow$)

Formal definitions

Consumption - also referred to as the personal consumption expenditure

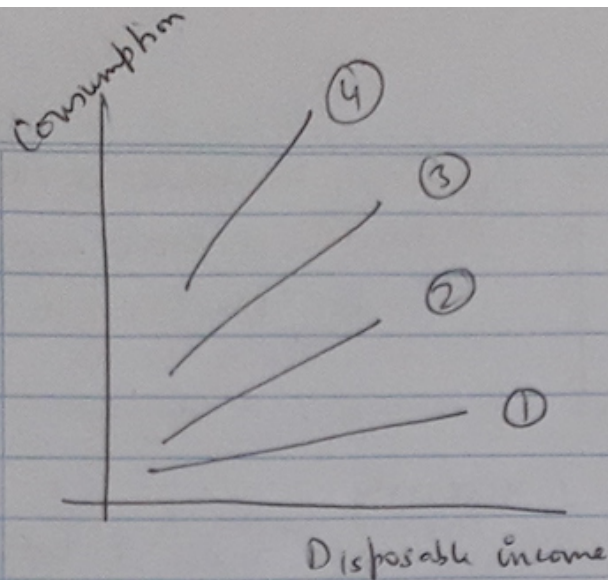
- is the expenditure by households on final goods & services.

Savings - part of disposable income that is not consumed.

$$\text{Disposable income} = \text{income} - \text{taxes}$$

How do consumption patterns differ by income groups?

- look at the figure on the left & the choices on the right. Please try to match the pairs.



③

- A - Housing
- B - Food
- C - Savings
- D - Medical care

[Answers & brief explanation on the next page]

- Long-term changes in consumption basket
 - How has our consumption basket changed over time?
 - A long term view for the US economy is presented below

Item(s)		At the time of First World War	Now (10, 100 years later)
1.	Food	41 %	14 %
2.	Transportation	1 %	11 %
3.	Recreation & entertainment	0 %	15 %

There were no mobiles / TVs etc at that time

these numbers indicate % of our consumption expenditure

A substantial chunk of comp. expenditure was devoted to food in that era. This is true of the poorest countries (or the poorest in developing country such as ours) in the recent times too.

- | | | |
|---|---|----------------|
| ① | - | B Food |
| ② | - | A Housing |
| ③ | - | D Savings |
| ④ | - | C Medical care |

Please try to relate it to the income elasticity concept we studied in micro.

food! - least inelastic / necessity

As income $\uparrow$, people (may) eat more & better quality food, but there is a limit on their expenditure on food

[In fact, As income $\uparrow$ the share of food expenditure $\downarrow$ this is referred to as the Engel's law in economics]

Recall Roti, Kapda Aur Makaan
[food, shelter & clothing]

Savings - People can afford to save only after their needs (including medical care) have been taken care of.

In fact, as we shall discuss subsequently, poor people may have negative savings

their spending $>$ income

poor may borrow to finance their needs

- o note that while we talk about the C, I etc or the ratios, we are considering the entire economy & not individuals

o Consumption, Savings & Income

The disposable income, by definition,

$$Y_d = \overset{\text{Income}}{Y} - \overset{\text{Taxes}}{T}$$

$$Y_d = \underset{\text{Consumption}}{C} + \underset{\text{Savings}}{S}$$

[∵ You can either consume or save]

Thus $Y = C + S + T$

We also studied $Y = C + I + G + NX$

for a closed economy [no transactions with Rest of world]

$$NX \equiv 0 \quad \text{or} \quad Y = C + I + G$$

Equating the two, $C + I + G = C + S + T$

If $G = T$ [i.e. the government budget is balanced or its

2) $I = S$

expenditure = tax revenue]

Investment that would be undertaken = Savings

This highlights the importance of savings

- o Consumption Function - shows the relationship between the level of consumption & the level of disposable income

↓
due to Keynes
↓

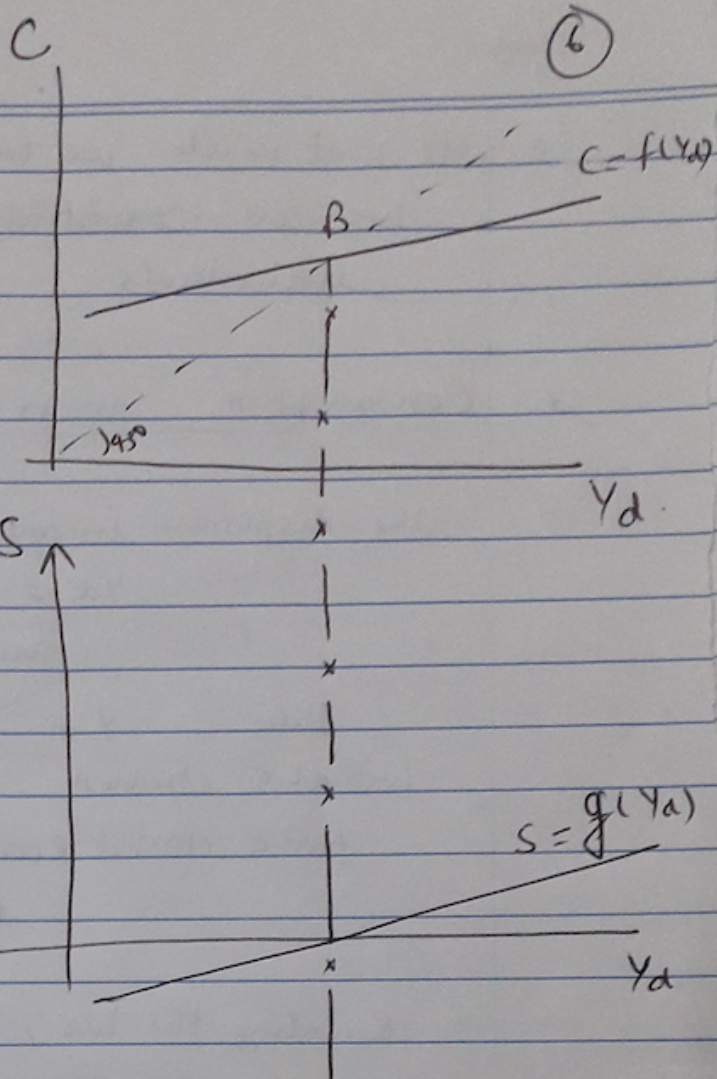
suggested a stable relationship between C & Y_d

$$C = f(Y_d)$$

+

(see the figure on the next page)

Break-even point (B) →
where disposable income
= Consumption expenditure



left of B $C > Y_d$ savings < 0
right of B $Y_d > C$ savings > 0

Savings function → relationship
between the level of savings
& the level of disposable
income

- mirror image of consumption
function

$$S = g(Y_d)$$

o Marginal Propensity to Consume (MPC)
How does consumption change with change in the level of
income

- is the slope of the consp. function

MPC: is the extra amount that people consume when they
receive an entire unit of income.

$$C(Y_d) = \alpha + \beta Y_d \quad \leftarrow \text{consp fn}$$

$$MPC = \frac{\partial C}{\partial Y_d} = \beta$$

Marginal Propensity to Save (MPS)

$$S(Y_d) = \gamma + \delta Y_d \quad \leftarrow \text{savings fn}$$

$$MPS = \frac{\partial S}{\partial Y_d} = \delta$$

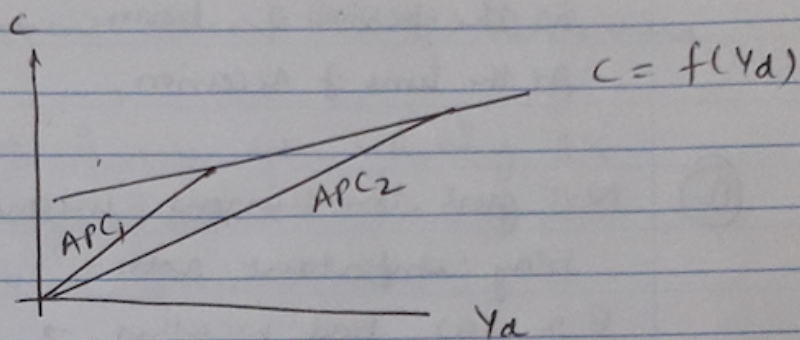
$$\therefore Y_d = C + S$$

$$\Rightarrow \boxed{\beta + \delta = 1} \text{ or } MPC + MPS = 1$$

• Average Propensity to Consume (APC) $= \frac{C}{Y_d}$

Note that

- 1) $APC > MPC$
- 2) while MPC is constant, APC varies with the level of Y_d
- 3) as $Y_d \uparrow$ $APC \downarrow$



($APC_2 < APC_1$) & approaches towards MPC

o National Consumption Behavior

Why are we interested in studying?

- $\therefore$ of its importance for business cycles & economic growth

Short-run

Long run

(See the first set of slides on Macro)

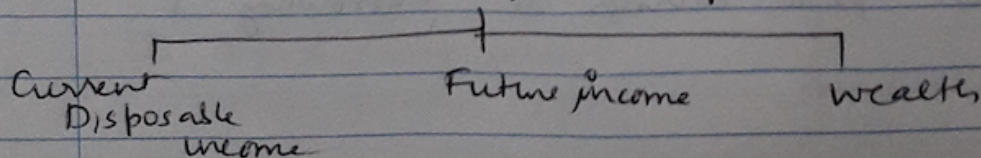
Short-run:- Consumption is a major part of aggregate spending & helps explaining business cycles

Long-run:- What is not consumed $\rightarrow$ savings $\xrightarrow{\text{(aids)}}$ investment $\rightarrow$ long-run growth

o Determinants of consumption

(Note that like all the concepts in macro, this is also aggregate consumption & NOT individual consumption)

Based on theory, three factors



(I) Disposable Income (Y_d)

changes in ^{Consumption} disposable income track the changes in disposable income

$$C = f(Y_d)$$

+

In the periods of boom, $C \uparrow$ as $Y_d \uparrow$

At the time of recession, $C \downarrow$ as $Y_d \downarrow$ usually.

(II) Not just the current income as in (I) but other factors too play important role,

E.g., (a) bad weather $\rightarrow$ if, in any particular year, bad weather (such as draught or floods) destroys crops, how do farmers survive?

$\rightarrow$ Farmers either borrow or rely on their past savings.

(b) Consider a student, who wishes to go abroad to pursue higher studies. How does s/he finance higher study? Education loan?

$\rightarrow$ s/he expects future income & borrows.

(c) Consider an individual engaged in any non-essential business activity, which had to be shut for say 2-3 months because of the ongoing covid-19 pandemic. There is no income for these 2-3 months. But the consumption of such a person does not $\rightarrow 0$ (zero), ~~but~~ This individual will rely on past savings or borrowing!

There are two theories that explain these situations

II.1 Permanent Income Hypothesis (PIH) →

due to Friedman

PIH says that not just the current income but permanent income (PI) determines consumption.

PI → is the trend level of income after removing the temporary or transient fluctuations / influences
[recall the concept of potential GDP]

People change their consumption if PI changes.

An important consequence of PIH is that one time bonus is unlikely to affect the level of consumption (unless such a bonus affects the PI or trend level of income)

II.2 Life Cycle Hypothesis (LCH) →

due to Modigliani

↳ says that individuals tend to smoothen their consumption over their life time income

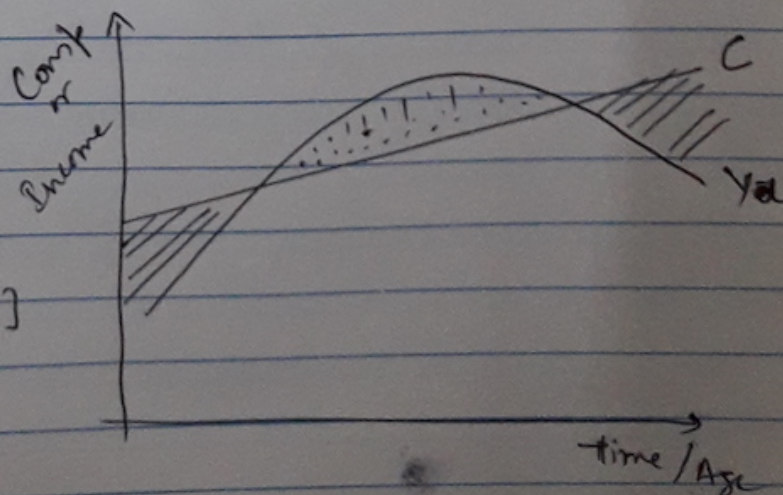
- people smoothen their consumption over time

C → Consumption

Y_d → disposable income

// → denotes the age groups / time period when $C > Y_d$ [savings < 0]

∴ → period / time interval when savings > 0



(of C & Y_d)

NOTE the shapes are tentative [in the sense that they are as observed for (a) developed world, (b) average for all individuals i.e. may not be true for each individual but will hold for economy as a whole]

- people tend to save a lot in the middle stage of their career, savings are usually at peak then, objective is to have adequate retirement income

III

Wealth

$$C_{comp} = f(\text{wealth})$$

Consider two families with same income but different levels of wealth.

[Difference between income & wealth ?]

Who do you think, A or B, will have more consumption, keeping all other things the same?

	A	B
Income	Rs 50K per month	Rs 50K per month
Wealth	Rs 50 lakhs	—

In 1929, at the time of Great Depression,

stock market crash $\rightarrow$ wealth $\downarrow \rightarrow$ consp $\downarrow$

Thus,

$$\text{Consumption} = f(\underbrace{Y_d}_{\substack{\text{Disposable} \\ \text{income}}}, \underbrace{Y_f}_{\substack{\text{future income} \\ \swarrow \searrow \\ \text{LCH} \quad \text{PIH}}}, \underbrace{W}_{\text{wealth}})$$

Investment

- Another major component of O/p
- In macroeconomics, investment does not mean financial investment but is usually the expenditure on capital goods or real assets:

↓
is something that is used to produce some other goods

When we talk about capital, we also distinguish between fixed capital vs working capital

↓
does not get used up immediately in the production process such as plant & machinery

↘ raw material (day-to-day inputs etc)

- Investment plays two roles -

(i) short-run :- its a large & volatile components of O/p. & explains business cycles.

(ii) long-run :- Investment → ↑ in potential o/p
long-run economic growth

if we want high growth in the long-run, we need policies aimed at promoting investment

- Types of investment → when we talk of investment, several types are possible -

Our focus for subsequent discussion shall be

⇒

- ① Gross private domestic investment
- ② foreign investment
- ③ government investment
- ④ investment in human beings

9 (12)

Thus, when we talk about investment in macroeconomics we usually mean investment in plants & machinery & other equipments.

• Determinants of Investment

Why do businesses undertake investment?

- because they want profit & we know that

$$\text{profit} = \underbrace{\text{Revenue}}_{\text{I}} - \underbrace{\text{Cost}}_{\text{II}}$$

① Revenue - when will an investment bring additional revenue?

When there are more sales ↓

& Sales depend on output, $f(y)$ output

Thus, profit depends on revenue, which depends on sales & in turn on output.

② Costs - cost to business of undertaking various investments.
What are these?

To raise capital, businesses take loans & pay interest.

Interest amount depends on interest rate.

If cost of borrowing (viz., interest rate) ↑ ⇒ investment ↓

Thus, I (a) interest rate (r) ; $I = I(r)$

② (b) Another cost (which reduces businesses' profits) is corporate tax.

Thus, Corporate tax rate too is a determinant of investment!

High tax rate → erode profits

& Govt provide tax-breaks to corporates to fuel investment

③ Future expectations - Another factor that affects investment is the expectations of business about future ~~macroeconomic~~ macro-economic & other conditions.

Investment $\rightarrow$ gamble about future
 Hence future profit expectations are important & the factors which affect them expectations (such as macro conditions, business confidence, political uncertainty, etc.) also affect investment.

• Investment Demand Curve.

Economists are usually interested in examining the relationship between investment & interest-rate:

- (i) interest rate is an economic variable
- (ii) " " a major policy instrument
 (later, we shall have the central bank use it)

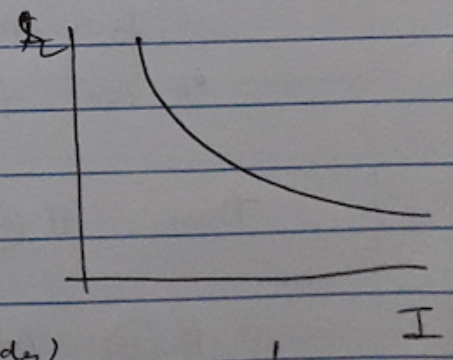
$\hookrightarrow$ Investment demand curve depicts the relationship between investment & interest-rate

$$I = I(r) \quad \ominus$$

Note that $r \rightarrow$ real interest rate

$$r = \underset{\substack{\swarrow \\ \text{nominal} \\ \text{interest rate}}}{i} - \underset{\substack{\searrow \\ \text{inflation}}}{\pi}$$

(see the earlier slides)



The investment demand schedule indicates the amount of investment that would be undertaken at each interest rate, r .

Shifts in Investment Demand

(See the previous discussion for a discussion on explanation)

(a) Output or GDP

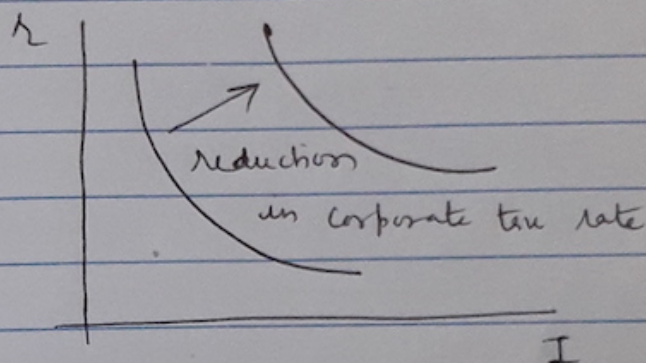
$$GDP_1 > GDP_0$$

higher $Y_p \Rightarrow$ higher capacity to buy goods $\Rightarrow$ revenue $\uparrow$



(b) Taxes (corporate)

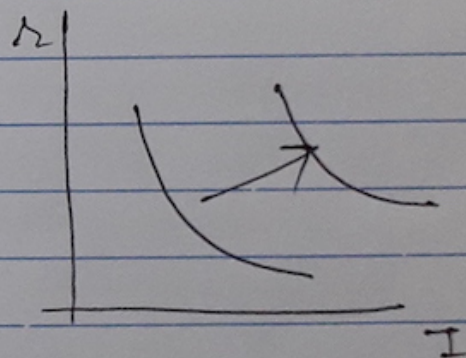
tax redⁿ $\Rightarrow I \uparrow$
(shifts out)



(c) business euphoria

- expectations about better future

e.g., 1990s $\rightarrow$ software companies & euphoria; but when the bubble busted in ~~1990s~~ 2000s, the demand for investment in software & equipment fell sharply & curve shifted left.



The term, Animal Spirits, is sometimes used to describe the optimism or pessimism of investors.

- Owing to various uncertainties, investment is the most volatile component of output.